

Instituto Tecnológico de Costa Rica

Operations Research - Semester II

Simplex

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Maximize

$$Z = 2.00x_1 - 1.00x_2 + 1.00x_3$$

Subject to

$$3.00x_1 + 1.00x_2 + 1.00x_3 \leq 60.00$$

$$1.00x_1 - 1.00x_2 + 2.00x_3 \leq 10.00$$

$$1.00x_1 + 1.00x_2 - 1.00x_3 \leq 20.00$$

Simplex Table

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	-2.00	1.00	-1.00	0.00	0.00	0.00	0.00
	0.00	3.00	1.00	1.00	1.00	0.00	0.00	60.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00

Intermediate Tables

Pivoting 1

Most Negative

Column 2 (-2.00)

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	-2.00	1.00	-1.00	0.00	0.00	0.00	0.00
	0.00	3.00	1.00	1.00	1.00	0.00	0.00	60.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00

Fractions

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b	Frac
	1.00	-2.00	1.00	-1.00	0.00	0.00	0.00	0.00	
	0.00	3.00	1.00	1.00	1.00	0.00	0.00	60.00	20.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00	20.00

$$60.00/3.00 = 20.00$$

$$10.00/1.00 = 10.00$$

$$20.00/1.00 = 20.00$$

Smallest fraction: 10.00 $\rightarrow$ Pivot: row 3

Pivot

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	-2.00	1.00	-1.00	0.00	0.00	0.00	0.00
	0.00	3.00	1.00	1.00	1.00	0.00	0.00	60.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00

Canonization

$$R_3 \leftarrow R_3/1.00$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	-2.00	1.00	-1.00	0.00	0.00	0.00	0.00
	0.00	3.00	1.00	1.00	1.00	0.00	0.00	60.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00

$$R_1 \leftarrow R_1 + 2.00R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	3.00	1.00	1.00	1.00	0.00	0.00	60.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00

$$R_2 \leftarrow R_2 + -3.00R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	1.00	1.00	-1.00	0.00	0.00	1.00	20.00

$$R_4 \leftarrow R_4 + -1.00R_3$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	2.00	-3.00	0.00	-1.00	1.00	10.00

Pivot Result

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	2.00	-3.00	0.00	-1.00	1.00	10.00

Pivoting 2

Most Negative

Column 3 (-1.00)

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	2.00	-3.00	0.00	-1.00	1.00	10.00

Fractions

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b	Frac
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00	
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00	7.50
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00	-10.00
	0.00	0.00	2.00	-3.00	0.00	-1.00	1.00	10.00	5.00

$$30.00/4.00 = 7.50$$

$$10.00/-1.00 = -10.00$$

$$10.00/2.00 = 5.00$$

Smallest fraction: 5.00 $\rightarrow$ Pivot: row 4

Pivot

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	2.00	-3.00	0.00	-1.00	1.00	10.00

Canonization

$$R_4 \leftarrow R_4/2.00$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	-1.00	3.00	0.00	2.00	0.00	20.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	1.00	-1.50	0.00	-0.50	0.50	5.00

$$R_1 \leftarrow R_1 + 1.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	0.00	1.50	0.00	1.50	0.50	25.00
	0.00	0.00	4.00	-5.00	1.00	-3.00	0.00	30.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	1.00	-1.50	0.00	-0.50	0.50	5.00

$$R_2 \leftarrow R_2 + -4.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	0.00	1.50	0.00	1.50	0.50	25.00
	0.00	0.00	0.00	1.00	1.00	-1.00	-2.00	10.00
	0.00	1.00	-1.00	2.00	0.00	1.00	0.00	10.00
	0.00	0.00	1.00	-1.50	0.00	-0.50	0.50	5.00

$$R_3 \leftarrow R_3 + 1.00R_4$$

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	0.00	1.50	0.00	1.50	0.50	25.00
	0.00	0.00	0.00	1.00	1.00	-1.00	-2.00	10.00
	0.00	1.00	0.00	0.50	0.00	0.50	0.50	15.00
	0.00	0.00	1.00	-1.50	0.00	-0.50	0.50	5.00

Pivot Result

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	0.00	1.50	0.00	1.50	0.50	25.00
	0.00	0.00	0.00	1.00	1.00	-1.00	-2.00	10.00
	0.00	1.00	0.00	0.50	0.00	0.50	0.50	15.00
	0.00	0.00	1.00	-1.50	0.00	-0.50	0.50	5.00

Results

	Z	x_1	x_2	x_3	s_1	s_2	s_3	b
	1.00	0.00	0.00	1.50	0.00	1.50	0.50	25.00
	0.00	0.00	0.00	1.00	1.00	-1.00	-2.00	10.00
	0.00	1.00	0.00	0.50	0.00	0.50	0.50	15.00
	0.00	0.00	1.00	-1.50	0.00	-0.50	0.50	5.00

Objective Value

$$Z = 25.00$$

Variables

$$x_1 = 15.00$$

$$x_2 = 5.00$$

$$x_3 = 0.00$$

Slack or Surplus

$$s_1 = 10.00$$

$$s_2 = 0.00$$

$$s_3 = 0.00$$